SWORD v17c Beta Release Notes

Version: v17c beta 0.0.4 **Date:** March 2026 **Authors:** James H. Gearon, Tamlin M. Pavelsky, Niek Collot d'Escury **Base version:** SWORD v17b (March 2025, UNC)

Changelog

0.0.4 (March 2026)

- **Node propagation to 11.1M nodes.** Five v17c columns (`best_headwater`, `best_outlet`, `pathlen_hw`, `pathlen_out`, `subnetwork_id`) were NULL on all nodes. Now propagated from parent reaches.
- **Dijkstra ghost outlet fix.** Ghost reaches (type=6) with out_degree=0 no longer report `dist_out_dijkstra=0`. All sinks are used as Dijkstra sources for full coverage (94–99% per region); ghost sinks receive NULL. Real outlet counts: NA=7, SA=1, EU=2, AF=1, AS=11, OC=1.
- **Bifurcation routing fix.** At multi-successor nodes, `rch_id_dn_main` now follows the mainstem chain unconditionally (was falling back to score-based ranking, which could disagree with `is_mainstem`). V023 (pathlen_out step consistency) violations: 2 → 0.
- **`hydro_dist_hw` computed.** Mainstem distance from headwater via `rch_id_up_main` chain walk (mirror of `hydro_dist_out`). Was stale from a prior pipeline run.
- **facc monotonicity fix (T003).** 408 reaches corrected via iterative downstream propagation (4 passes). T003 violations: 392 → 5.
- **Routing weights retrained on `effective_slope`.** SWOT `slope_obs_p50` (where reliable and n 5) replaces MERIT DEM slope in routing score training. Slope share: 8% → 3%, width: 67% → 71%. CV accuracy unchanged (88.5%).
- **Node-level distance interpolation.** `hydro_dist_out`, `hydro_dist_hw`, and `dist_out_dijkstra` added to nodes table, interpolated by node position within reach (using v17b `dist_out` offset). `pathlen_hw` and `pathlen_out` changed from flat reach copies to per-node interpolated values.
- **Variable reference updated.** 7 missing variables added, 8 type mismatches fixed, `cl_ids` shape corrected to `cl_id_min`/`cl_id_max`.

0.0.3 (March 2026)

- **Routing weights learned from human labels.** Replaced the handcrafted lexicographic 3-tuple `(effective_width, log_facc, pathlen)` with a weighted scalar score trained on 1,967 human-labeled junction decisions: ‘1.97*log1p(ew) + 0.23*log1p(facc) - 0.23*log1p(slope) + 0.23*log1p(pathlen)
  - 0.29*stream_order‘. Two new signals vs prior releases: slope (negative = prefer lower gradient) and stream_order. All routing functions use the same score to prevent divergence. (Weights retrained in 0.0.4; see above.)
- **Caroline's reviewer fixes synced.** 127 C001 lakeflag fixes (NA) and 10 C004 type fixes (NA) from the Streamlit reviewer app.
- **NA PostgreSQL geometry fix.** 38,696 NA reaches had NULL geometry in PostgreSQL — v17b geometry overwrite via dblink silently failed. Rewrote to read from v17b GPKG files on disk with verification gates.
- Fixed `dn_node_id`, `up_node_id`, and `node_order` for 810 flow-corrected reaches where node ordering was stale (based on v17b `dist_out`). 639 reaches now have `dn_node_id !=`

`min(node_id)`, reflecting the inverted flow direction.
- Removed redundant `rch_id_up_1..4` / `rch_id_dn_1..4` vector variables from NetCDF export. Only the `[4, N]` matrices `rch_id_up` and `rch_id_dn` are exported, matching v17b format.
- Fixed `swot_orbits` NetCDF type from string back to int64 (matching v17b).
- Fixed 15 integer columns widened from int32 to int64 in NetCDF export (`n_nodes`, `lakeflag`, `n_rch_up`, `stream_order`, etc.). All shared variable types now match v17b exactly.
- Synced `rch_id_up_1..4` / `rch_id_dn_1..4` DB columns from `reach_topology` table (694 were stale after flow corrections).

0.0.2 (March 2026)

- Renamed `is_mainstem_edge` to `is_mainstem`
- Mainstem algorithm refactored: computed per `main_path_id` group (see Section 2.1)
- Fixed `n_rch_up`/`n_rch_down` stale counts at 148 flow-corrected reaches
- Recomputed `facc` at 807 flow-corrected reaches using topological propagation; resolves all F006 junction conservation violations and T003 monotonicity violations at these reaches
- OC reach 51111300061 incomplete split fully reverted to v17b state (434 orphan centerlines, 73 orphan nodes restored)

0.0.1 (March 2026)

- Initial v17c beta release
- Includes `node_order`, `dn_node_id`, `up_node_id` (node ordering within reaches)

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1. Overview

SWORD v17c extends v17b with three additions: computed mainstem topology, SWOT observation statistics, and flow accumulation corrections. No reaches, nodes, or centerlines were added or removed. v17c contains the same 248,673 reaches, 11.1M nodes, and 66.9M centerline points as v17b across all six regions (NA, SA, EU, AF, AS, OC).

Each region is distributed as a single NetCDF4 file (`{region}_sword_v17c_beta.nc`). The group structure matches v17b (centerlines, nodes, reaches), and the `area_fits` and `discharge_models` subgroups under reaches pass through from v17b unchanged. Reach and node ordering within each file matches v17b.

Reach coordinate columns (`x`, `y`, `x_min`, `x_max`, `y_min`, `y_max`) match v17b values across all formats (NetCDF, DuckDB, PostgreSQL).

All new variables use a fill value of -9999 where no observation or computation produced a value.

For a complete variable catalog, see v17c_variable_reference.md.

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2. New Variables

2.1 Mainstem Topology (reaches group)

`is_mainstem` is computed per `main_path_id` group: each group (a mainstem path plus its tributary branches) gets one canonical chain, identified by a greedy walk from the group's shared `best_headwater`. At each junction the algorithm selects the upstream branch with the highest weighted routing score: `2.02*log1p(ew) + 0.17*log1p(facc) - 0.08*log1p(slope) + 0.35*log1p(pathlen) + 0.23*stream_order`. These weights were learned from 1,967 human-labeled junction decisions via logistic regression on pairwise log1p-difference features, using SWOT slope where reliable (else MERIT DEM). The negative slope weight captures the geomorphic pattern that mainstem channels have lower gradients than tributaries. Mainstem reaches within each group are then assigned `rch_id_up_main` / `rch_id_dn_main` from the chain; non-mainstem reaches use the same weighted score. Ghost reaches (type=6) are excluded from mainstem but still participate in routing topology. ~10% of mainstem reaches have no mainstem neighbor at `main_path_id` group boundaries — this is expected by design.

Variable	Type	Units	Description
`dist_out_dijkstra`	float64	meters	Dijkstra shortest-path distance from the upstream end of the reach to any network outlet (outlet = 0; NULL for ghost reaches)
`hydro_dist_out`	float64	meters	Mainstem distance from the upstream end of the reach to `best_outlet` via `rch_id_dn_main` chain (outlet = `reach_length`)
`hydro_dist_hw`	float64	meters	Mainstem distance from `best_headwater` to the downstream end of the reach via `rch_id_up_main` chain (headwater = `reach_length`)
`rch_id_up_main`	int64	—	Main upstream neighbor reach_id (mainstem-preferred)
`rch_id_dn_main`	int64	—	Main downstream neighbor reach_id (mainstem-preferred)
`best_headwater`	int64	—	Routing-score-prioritized headwater reach_id for the network component
`best_outlet`	int64	—	Routing-score-prioritized outlet reach_id for the network component
`pathlen_hw`	float64	meters	Cumulative path length from `best_headwater`
`pathlen_out`	float64	meters	Cumulative path length to `best_outlet`

Variable	Type	Units	Description
is_mainstem	int32	—	1 if reach is on a mainstem path, 0 otherwise
main_path_id	int64	—	Unique identifier for each mainstem path group
subnetwork_id	int32	—	Connected component ID (Pfafstetter-offset, globally unique; see Section 4)
dn_node_id	int64	—	Node ID at the downstream end of the reach (lowest dist_out)
up_node_id	int64	—	Node ID at the upstream end of the reach (highest dist_out)

Eight of these variables also appear at node level. Five are interpolated by node position within the reach: hydro_dist_out, hydro_dist_hw, dist_out_dijkstra, pathlen_hw, and pathlen_out. Three are flat copies from the parent reach: subnetwork_id, best_headwater, and best_outlet. For flow-corrected reaches (660), the interpolation reverses the offset direction to produce correct monotonic distances.

The interpolation offset for each node is reach.dist_out - node.dist_out, where node.dist_out is the v17b original (not recomputed). This offset equals 0 at the upstream end of the reach and reach_length at the downstream end. For single-node reaches the offset is reach_length / 2 (midpoint). Node-level dist_out itself is the v17b value, preserved as-is for backward compatibility; at single-node reaches it equals the reach-level value rather than the midpoint.

At reach boundaries the downstream-end node of the upstream reach and the upstream-end node of the downstream reach share identical interpolated distance values (verified gap = 0 m across all boundaries).

node_order is a node-level variable (not in the reaches table): 1-based position within a reach, ordered by dist_out ascending (1 = downstream end, n = upstream end).

Distance convention note. dist_out (v17b) and hydro_dist_out both measure from the upstream end of the reach and assign reach_length at the outlet. dist_out_dijkstra also measures from the upstream end but assigns 0 at the outlet; the offset between hydro_dist_out and dist_out_dijkstra at any reach equals the outlet's reach_length. hydro_dist_hw measures from the headwater to the downstream end of the reach. See the variable reference for the full convention table.

2.2 SWOT Observation Statistics

Percentile-based summaries computed from available SWOT observations. All percentile, range, and MAD variables share the units of the underlying measurement.

Reaches and nodes:

Variable	Type	Units	Description
`wse_obs_p10`–`wse_obs_p90`	float64	meters	WSE percentiles (10th through 90th, in steps of 10)
`wse_obs_range`	float64	meters	WSE observation range (p90 - p10)
`wse_obs_mad`	float64	meters	WSE median absolute deviation
`width_obs_p10`–`width_obs_p90`	float64	meters	Width percentiles
`width_obs_range`	float64	meters	Width observation range
`width_obs_mad`	float64	meters	Width median absolute deviation
`n_obs`	int32	—	Total SWOT observation count

Reaches only:

Variable	Type	Units	Description
`slope_obs_p10`–`slope_obs_p90`	float64	m/km	Slope percentiles
`slope_obs_range`	float64	m/km	Slope observation range
`slope_obs_mad`	float64	m/km	Slope median absolute deviation
`slope_obs_adj`	float64	m/km	Adjusted slope
`slope_obs_slopeF`	float64	—	Slope F-statistic
`slope_obs_reliable`	int32	—	0 = unreliable, 1 = reliable
`slope_obs_quality`	int32	—	Integer quality category (0–8; see Section 3)
`slope_obs_n`	int32	—	Number of slope observations
`slope_obs_n_passes`	int32	—	Number of SWOT passes used
`slope_obs_q`	int32	—	Bitfield quality flag (see Section 3)

2.3 Flow Accumulation Corrections

A two-stage denoise pipeline corrected flow accumulation (`facc`) values to address three systematic error modes in MERIT Hydro's D8 (eight-direction flow routing) upstream area: bifurcation cloning, junction inflation, and raster-vector misalignment. The pipeline corrected 96,589 of 248,673 reaches (38.8%). Uncorrected reaches retain v17b values. See facc_correction_methodology.md for the full algorithm description.

Variable	Type	Group	Description
`facc`	float64	reaches, nodes	Flow accumulation (km^2). Corrected values where applicable; v17b values otherwise.
`facc_quality`	int32	reaches, nodes	1 = corrected by denoise_v3; fill_value = not flagged

After correction, junction conservation violations (downstream facc < sum of upstream facc) are resolved in all regions. In 0.0.2, facc was additionally recomputed at 807 flow-corrected reaches via topological propagation, resolving remaining monotonicity violations at those reaches.

2.4 Other New or Updated Variables

Variable	Type	Group	Description
`type`	int32	reaches	Reach classification (1=river, 3=lake_on_river, 4=dam, 5=unreliable, 6=ghost). Not present in v17b NetCDF; added in v17c so NetCDF users can filter by reach type without needing the database.
`dl_grod_id`	int64	reaches	DL-GROD (Deep Learning Global River Obstruction Database; He et al. 2025) dam/obstruction ID
`edit_flag`	string	reaches	Tag for manually edited reaches (e.g., `lake_sandwich`, `harp_lake`)

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3. Flag Encoding Reference

facc_quality

Value	Meaning
1	`denoise_v3` — corrected by the facc denoising pipeline
-9999 (fill)	Not flagged; facc unchanged from v17b

CF attributes: `flag_values = [1]`, `flag_meanings = "denoise_v3"`.

slope_obs_quality

Value	Meaning
0	reliable
1	small_negative
2	moderate_negative
3	large_negative
4	negative
5	below_ref_uncertainty
6	high_uncertainty
7	noise_high_nobs

Value	Meaning
8	flat_water_noise

CF attributes: `flag_values = [0,1,2,3,4,5,6,7,8]`, `flag_meanings = "reliable small_negative moderate_negative large_negative negative below_ref_uncertainty high_uncertainty noise_high_nobs flat_water_noise"`.

slope_obs_reliable

Value	Meaning
0	Unreliable
1	Reliable
-9999 (fill)	Not computed (no SWOT slope observations)

slope_obs_q (bitfield)

Bit	Value	Meaning
1	1	Negative slope
2	2	Low number of passes
3	4	High variance
4	8	Extreme value
5	16	Clipped

Flags combine by addition. A value of 0 indicates no quality issues. Example: 5 = negative slope (1) + high variance (4).

is_mainstem

Value	Meaning
0	Not on mainstem
1	On mainstem path

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4. Known Limitations

- **SWOT observation coverage:** SWOT statistics are fill_value (-9999) for reaches and nodes lacking SWOT data.
- **facc correction scope:** 95,913 reaches corrected (38.6%); the remaining 152,761 retain v17b values. Node-level facc propagates from the parent reach.
- **Lake sandwich corrections:** 1,252 reaches reclassified to `lakeflag = 1` where a narrow, shorter-than-neighbor reach sat between lake reaches (tagged `edit_flag = "lake_sandwich"`). ~1,755 similar cases remain (narrow connecting channels, chains).

- **HarP lake corrections:** 7,425 reaches reclassified from `lakeflag = 0` (river) to `lakeflag = 1` (lake) based on HarP v1.1 (Hydrography and River Planform) lake classification data. 200,201 child nodes updated to match. Tagged `edit_flag = "harp_lake"`. Existing tags preserved (comma-delimited when multiple apply).
- **area_fits and discharge_models:** Direct copies from v17b. Not recomputed against v17c facc or SWOT values.
- **`subnetwork_id` vs `network`:** `subnetwork_id` uses Pfafstetter- offset enumeration (globally unique). v17b `network` uses per-region 1-based IDs. Different component counts (v17c finds more via weakly connected components; 19 subnetworks span multiple v17b networks). `network` is retained unchanged from v17b.
- **Topology reciprocity gaps (resolved):** 151 non-reciprocal pairs (both reaches listing each other as upstream) were introduced during flow correction revert. Fixed by completing the revert for all affected reciprocal entries. Zero non-reciprocal pairs remain.
- **Flow correction oscillation:** 389 reaches (0.16%) in AF/AS/EU/NA/SA had ambiguous WSE slope signals causing bidirectional flow correction scores. These were reverted to v17b topology.
- **main_path_id consistency:** 3,134 reaches have `main_path_id` values inconsistent with current `(best_headwater, best_outlet)` tuples (V013- V015 lint checks). 80 reaches in NA have `best_headwater` pointing to non-headwater reaches. Requires recomputing `main_path_id` from current headwater/outlet assignments.
- **River naming:** 51.2% of reaches are unnamed (NODATA), ranging from 26% (AF) to 69% (OC). 2.6% of mainstem 1:1 links have local name discontinuities (name changes between adjacent reaches with no junction).
- **Width fill values (A003):** ~1,266 reaches have `width=0` (unmeasured) or `width=-1` (GRWL lake fill). These are v17b fill values, not data errors. Present in both v17b and v17c unchanged. The A003 lint check is downgraded to WARNING for this reason.
- **`lakeflag/type` mismatch:** ~5,770 reaches have `lakeflag=1` (lake) but `type=1` (river), introduced by HarP and lake-sandwich corrections which updated `lakeflag` but not `type`. `type` is encoded in the last digit of `reach_id`, so changing it would change reach IDs. Policy for reach_id changes under discussion. Affects users filtering on both `lakeflag` and `type` at river-lake boundaries. Tracked in issue #208.

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5. Quality Audits

Validation checks performed on the v17c data:

Audit	Finding
Geometry	DuckDB geometries (rebuilt from NetCDF) lack endpoint overlap vertices present in v17b (210,533 reaches affected: 173K +1 point, 37K +2 points). `reach_length` unchanged. Reach coordinate columns (`x`, `y`, `x_min`, `x_max`, `y_min`, `y_max`) copied from v17b to ensure consistency across all formats.
n_nodes / reach_length	Internally consistent. Zero N008/G002/G003 violations.
path_freq gaps	v17b had 4,952 connected non-ghost reaches with invalid path_freq (0 or -9999). Resolved in v17c; remaining nodata values are correctly attributed to ghost reaches (type=6).
subnetwork_id	3,027 components across 248,673 reaches verified. Pfafstetter banding correct. Zero cross-region collisions. 19 subnetworks (0.6%) span multiple v17b networks (expected).
Topology integrity	T001 (dist_out_dijkstra monotonicity), T012 (referential integrity), T013 (self-reference), T014 (bidirectional): all pass. T005/T007: zero non-reciprocal edges (151 from incomplete flow correction revert resolved in beta 0.0.1). Note: v17b `dist_out` is stale at 807 flow-corrected reaches where topology direction was updated but `dist_out` retains its v17b value — use `dist_out_dijkstra` or `hydro_dist_out` for distance routing.
n_rch_up/n_rch_down	148 scalar count mismatches corrected (flow corrections flipped reach_topology but did not recalculate counts). Zero mismatches across all 248,673 reaches.
OC reach split revert	Incomplete `break_reaches()` split of OC reach 51111300061 (434 orphan centerlines, 73 orphan nodes) fully reverted to v17b state.
River name formatting	291 formatting issues corrected (separators, whitespace). Automated checks now enforce ";" separator and alphabetical ordering.
Flow direction	1,112 experimental topology flips reverted after causing 30K disconnected reaches. 807 reaches across all regions retain corrected topology that differs from v17b (175 sections, median 5 reaches/section). OC has 119 of these (26 SWOT-validated sections). Non-OC corrections were retained from the flow correction pipeline; `dist_out` is stale at these reaches — use `dist_out_dijkstra`.

HarP lake corrections	7,425 reaches reclassified lakeflag 0 to 1 from HarP v1.1 data. 200,201 nodes propagated. Tagged `edit_flag = "harp_lake"`.

For POM (Pierre-Olivier Malaterre) validation results, see pom_validation_report.md.

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6. File Format

- **Format:** NetCDF4 (one file per region)
- **Naming:** `{region}_sword_v17c_beta.nc` where region is `na`, `sa`, `eu`, `af`, `as`, `oc`
- **Groups:** `centerlines`, `nodes`, `reaches`
  - `reaches/area_fits` and `reaches/discharge_models` subgroups (from v17b)
- **Ordering:** Reach, node, and centerline arrays match v17b ordering
- **Fill value:** -9999 for all numeric variables (int32, int64, float64)
- **Checksums:** SHA256 hashes listed in `SHA256SUMS_{version}.txt`
- **Additional formats:** GeoPackage and GeoParquet exports available (reaches and nodes per region, with geometry)

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7. Methodology Documentation

Document	Description
facc_correction_methodology.md	Facc denoise algorithm, detection rules, correction model
pom_requests_summary.md	POM validation check tracker (19 checks, production results)
pom_validation_report.md	POM validation production results
v17c_variable_reference.md	Complete variable catalog for NetCDF export
SWORD_v17b_Technical_Documentation.md	v17b baseline reference